

# Robert Lilly

## Mechanical Engineering Student

University of Connecticut Honors Program

Mechanical Engineering student in the University of Connecticut Honors Program with interests in robotics, mechatronics, control systems, and embedded technologies. Actively involved in engineering organizations and hands-on technical projects focused on engineering analysis and intelligent systems.

### TECHNICAL PROJECTS & ENGINEERING EXPERIENCE

#### Robotics & Engineering Software Projects — Python/MATLAB

- Developed engineering-analysis and robotics tools involving inverse kinematics, control-system response analysis, beam reaction calculations, and stress-strain modeling
- Created robotics kinematics utilities for robotic arm motion analysis and workspace calculations
- Built engineering-focused software tools emphasizing numerical analysis, mechanics, and problem solving
- Published engineering and robotics-related software projects through a public GitHub portfolio

#### Arduino Ultrasonic Sensor Project

- Designed an Arduino-based ultrasonic sensing system with LED feedback for obstacle and distance detection
- Integrated ultrasonic sensors, LEDs, and Arduino microcontrollers using basic circuit design principles
- Developed embedded hardware architecture and published all source code to a public GitHub repository

#### Renewable Energy Proposal Initiative — University of Connecticut

- Developed a renewable energy proposal for UConn Stamford focused on solar infrastructure, solar canopies, and sustainability initiatives
- Submitted proposal to UConn President Radenka Maric as part of independent engineering and sustainability research

#### Engineering Simulation Experience

- Completed coursework and technical modeling projects using ANSYS Workbench for engineering analysis and simulation workflows

#### Leadership & Mentoring Experience

##### *Robotics and Beyond — Camp Assistant*

- Mentored younger students during robotics-focused summer programs and activities

##### *Legacy Martial Arts — Assistant Sensei*

- Assisted students during classes while helping maintain a structured and supportive learning environment
- Demonstrated leadership, discipline, and communication skills through mentorship and instruction

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### EDUCATION

#### University of Connecticut

Bachelor of Science in  
Mechanical Engineering  
University Honors Program  
Expected Graduation: 2028  
GPA: 3.555

**Focus Areas:** Robotics,  
Mechatronics, Control  
Systems, Embedded Systems

### TECHNICAL EXPERIENCE

- Python
- MATLAB
- SolidWorks
- CAD Modeling
- Arduino
- ANSYS Workbench
- Engineering Analysis
- Control Systems

### ENGINEERING ORGANIZATIONS

- FIRST, at UConn
- Underwater Robotics Team at UConn
- FROST Robotics Club
- VEX Robotics Club

### AWARDS & CERTIFICATIONS

- Dean's List — Spring 2025
- IBM AI Fundamentals Certification
- Teacher's Appreciation Scholarship — 2024
- 3rd Place — Southern New England VEX Championship (2024)
- First Degree Black Belt in Kempo Karate